

G.G.D. SANUSHKA DIAS

B.Sc. (Honours) in Computer Science

University of Sri Jayewardenepura

 [GitHub](#) |  [LinkedIn](#) |  [Portfolio](#) |  [Gmail](#) |  [WhatsApp](#)

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1 PROFESSIONAL PROFILE

Passionate and research-oriented 4th year undergraduate with strong interests in **Machine Learning Operations (MLOps)**, **Large Language Models (LLMs)**, **AI Systems**.

Experienced in developing AI-powered systems, full-stack applications, automation workflows, and research-focused machine learning solutions.

Skilled in Python development, Docker-based environments, model experimentation, DevOps concepts, and scalable AI workflows. Actively exploring advanced topics including neural network unlearning, LLM fine-tuning, model optimization, monitoring, deployment pipelines, and AI safety. Strong problem-solving abilities with hands-on experience building practical intelligent systems.

2 EDUCATION

University of Sri Jayewardenepura, Colombo, Sri Lanka 2023 – Present

B.Sc. (Honours) in Computer Science

Major Subjects: Computer Science | Mathematics | Physics

Currently in 4th Year Undergraduate Studies

Ananda College, Colombo 10, Sri Lanka 2021

G.C.E. Advanced Level (Physical Science Stream)

Chemistry: B | Physics: B | Combined Mathematics: C

Ananda College, Colombo 10, Sri Lanka 2017

G.C.E. Ordinary Level

6 A's and 3 B's

3 PROFESSIONAL TRAINING & CERTIFICATIONS

Diploma Level - iCET Certified Developer Institute of Computer Engineering Technology

Completed diploma-level certification covering software development fundamentals, programming practices, and application development.

4 RESEARCH & TECHNICAL PROJECTS

1. Full-Parameter Selective Unlearning Research for LLMs (Ongoing)

- Conducting advanced research on removing toxic knowledge from already trained Large Language Models using full-parameter selective unlearning while preserving useful knowledge and general model performance.
- Exploring methodology including forget, retain, and control sets, stability regularization, gradient-based full-parameter updates, and evaluation using established international benchmarks such as RealToxicityPrompts, ToxiGen, MUSE, and BOLD.
- Tools & Technologies: Python, PyTorch, Hugging Face, CUDA, Docker, Pandas, NumPy, Matplotlib

2. Full-Weight Harmful Knowledge Unlearning for 1.5B LLM

- Research on selective harmful knowledge reduction in a 1.5B Large Language Model using full-weight WMDP-targeted unlearning.
- Updated real model weights, including selected transformer layers, final normalization layer, and LM head, while preserving safe utility through retain objectives.
- Evaluated harmful knowledge reduction using WMDP-Bio, WMDP-Cyber, WMDP-Chem, Safe MCQ, retain prompts, and robustness checks.
- Tools & Technologies: Python, PyTorch, Hugging Face Transformers, CUDA, WMDP Benchmark, LLM Evaluation

[GitHub Repository](#)

3. Adapter-Based Safety Exploration for 9B LLM

- Explored adapter-based safety and unlearning strategies on a 9B Large Language Model to study harmful output suppression and model behaviour under safety constraints.
- Tested lightweight intervention methods for reducing unsafe responses while analysing limitations compared with direct full-weight knowledge removal.
- Used the 9B experiments as an exploratory research track to understand scalability, refusal behaviour, and robustness evaluation in larger models.
- Tools & Technologies: Python, PyTorch, Hugging Face Transformers, PEFT, LoRA, CUDA, LLM Safety Evaluation

[GitHub Repository](#)

4. AI-Based Trip Advisor Robot (Project Sasha)

- Developed an intelligent robotic travel assistant capable of personalized travel recommendations, speech interaction, face tracking, and physical movement control.
- Integrated computer vision, speech recognition, and AI-based interaction systems for real-time user engagement.
- Tools & Technologies: Python, Arduino, ESP32, OpenAI API, Webcam, OLED Displays

[GitHub Repository](#) | [Demo Video on YouTube](#)

5. Gift A Future – Child Donation Platform

- Developed a full-stack donation platform supporting education and welfare initiatives for children in need.
- Implemented secure workflows, database management, user authentication, and responsive interfaces.
- Tools & Technologies: React, Spring Boot, MySQL, Docker

[GitHub Repository](#)

6. Restaurant Table Management System

- Overview: Real-time desktop app for table reservations and seating allocation.

- Tools & Technology: Java Swing, MySQL
[GitHub Repository](#)

5 TECHNICAL SKILLS

- **Programming Languages:** Python, Java, JavaScript, PHP, HTML, CSS
- **Machine Learning & AI:** PyTorch, Hugging Face Transformers, LLM Fine-Tuning, Neural Networks, Deep Learning Concepts
- **MLOps & DevOps:** Docker, CI/CD Concepts, Monitoring Concepts, Model Deployment, Git, GitHub
- **Frameworks & Tools:** React, Spring Boot, VS Code, IntelliJ IDEA
- **Databases:** MySQL
- **Research Areas:** Selective Machine Unlearning, AI Safety, LLM Optimization, Deep Learning
- **Soft Skills:** Research Thinking, Problem Solving, Teamwork, Communication, Fast Learning

6 WORK EXPERIENCE

Customer Relation Officer

Nov 2022 – May 2023

Kangaroo Cabs (Pvt) Ltd, Sri Lanka

- Managed inbound customer calls and provided timely support.
- Collected and maintained accurate customer information.
- Handled customer complaints professionally and ensured effective resolution.
- Developed strong communication, problem-solving, and interpersonal skills.
- Worked collaboratively as a team player while maintaining responsibility and discipline.

7 LANGUAGES

English (Professional Working Proficiency) | Sinhala (Native)

8 REFERENCES

Prof. R.G.N. Meegama

Senior Professor in Computer Science

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